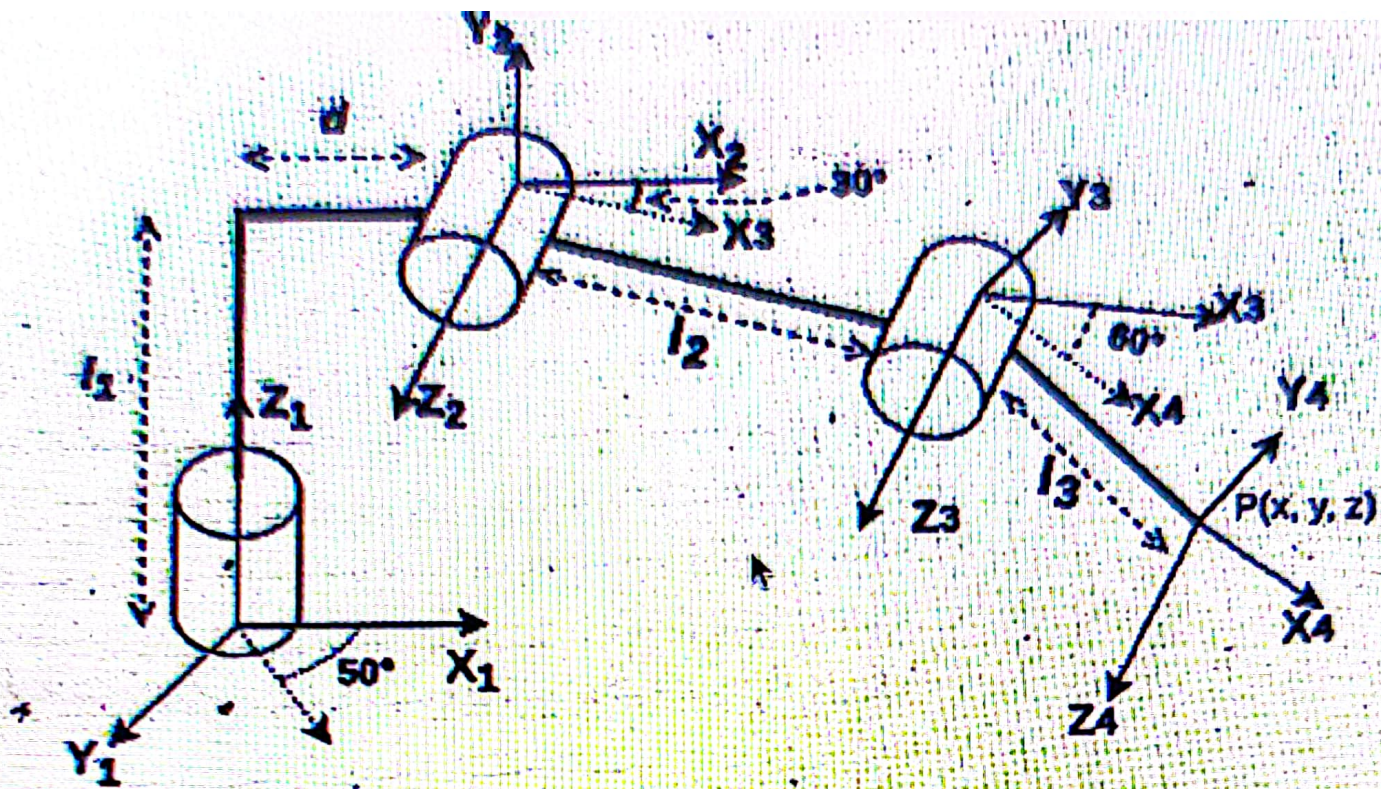
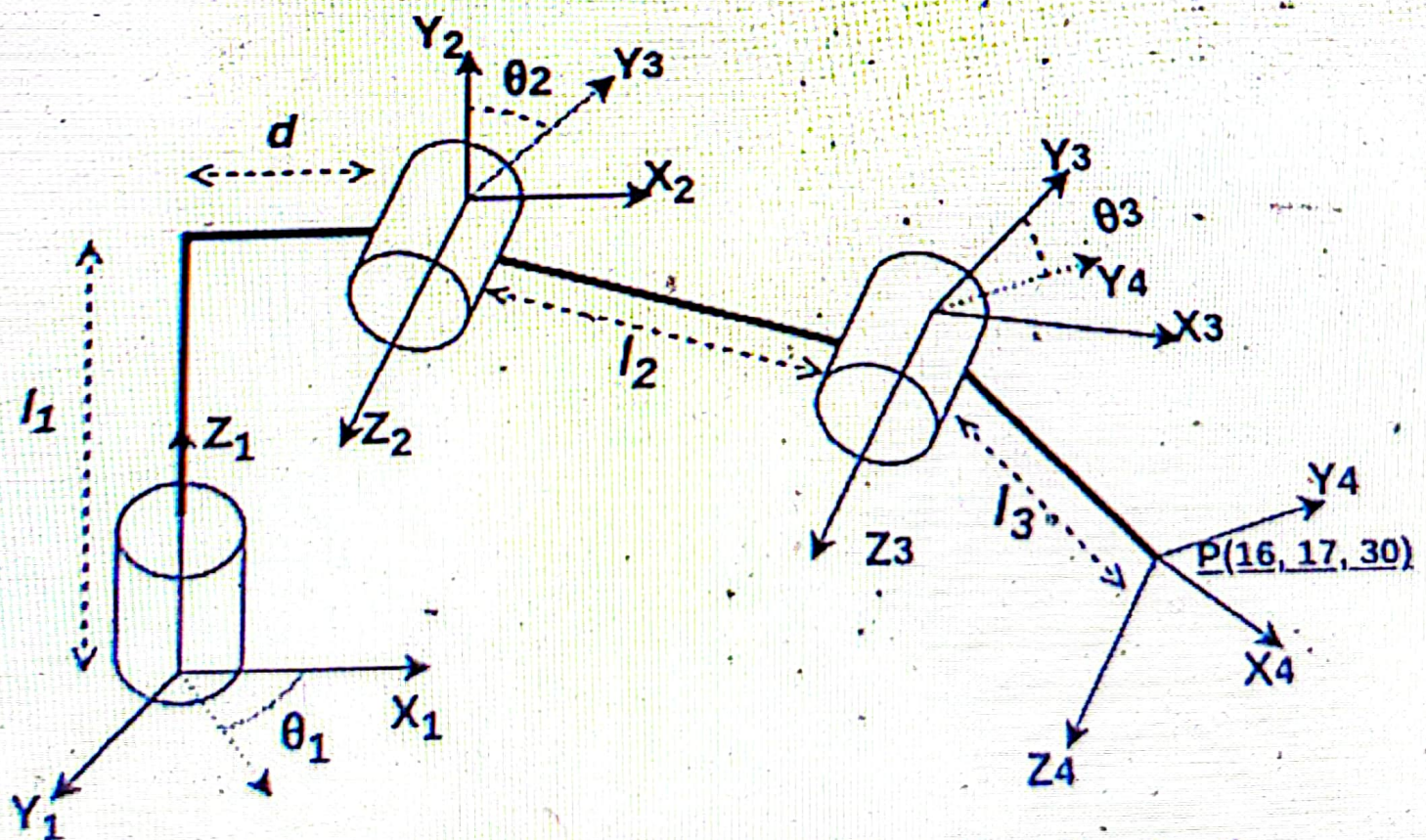




Given: $l_1 = 15$, $d = 3$, $l_2 = 18$, $l_3 = 16$. Find $P(x, y, z)$.

Question 2:

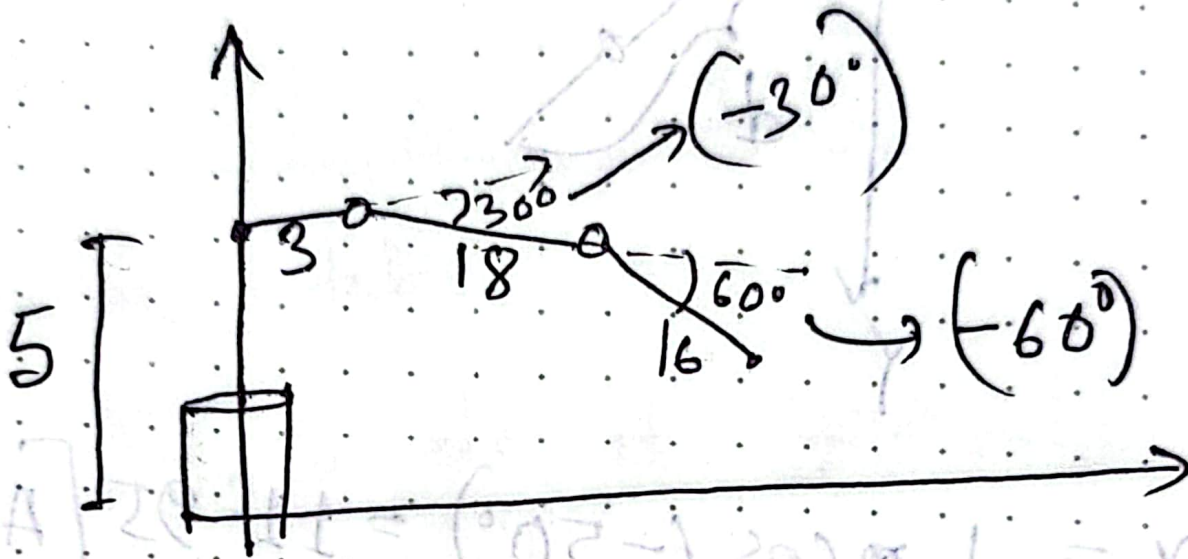


Given: $l_1 = 18$, $d = 2$, $l_2 = 20$, $l_3 = 14$. Find θ_1 , θ_2 , θ_3 .

Q1/

Trigonometry approach

side View:



$$d = 3 + 18 \cdot \cos(-30^\circ) + 16 \cdot \cos(-30^\circ - 60^\circ)$$

$$= 18.589$$

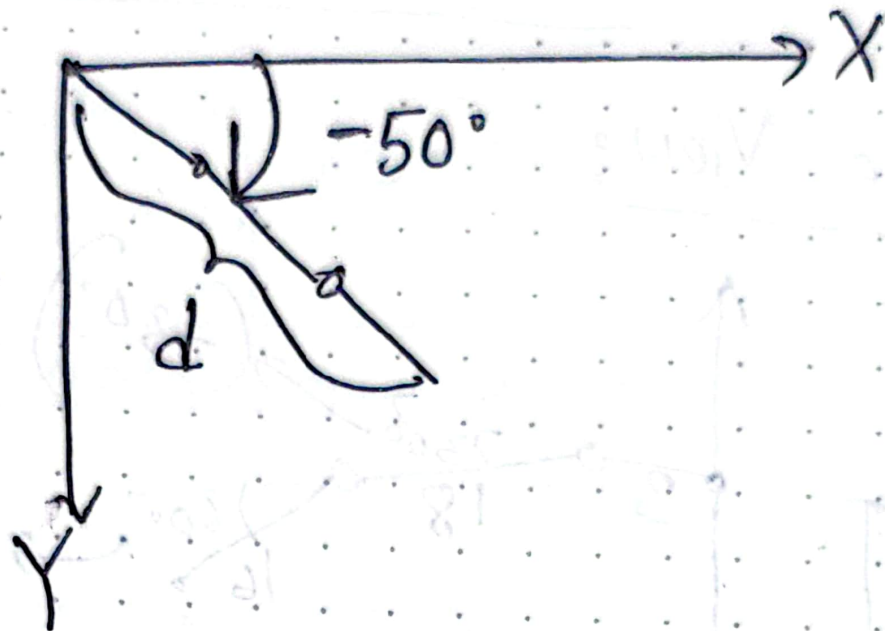
$$z = 15 + 18 \cdot \sin(-30^\circ) + 16 \cdot \sin(-30^\circ - 60^\circ)$$

$$= -10$$

Date: / /



Top view:



$$x = d \cdot \cos(-50^\circ) = 11.95 \text{ [Ans]}$$

$$y = d \cdot \sin(-50^\circ) = -14.24 \text{ [Ans]}$$

DH

i	θ	α	a	d
1	-50°	00°	3(d)	(1) 15
2	-30°	0	18(1)	0
3	-60°	0	16(1)	0

From the table,

${}^1T_2 = \text{do it yourself}$

${}^2T_3 =$

${}^3T_4 =$

$${}^1T_4 = \begin{bmatrix} 0 & 0.6428 & -0.76 & 11.95 \\ 0 & -0.766 & -0.64 & -14.23 \\ -1 & 0 & 0 & -10 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

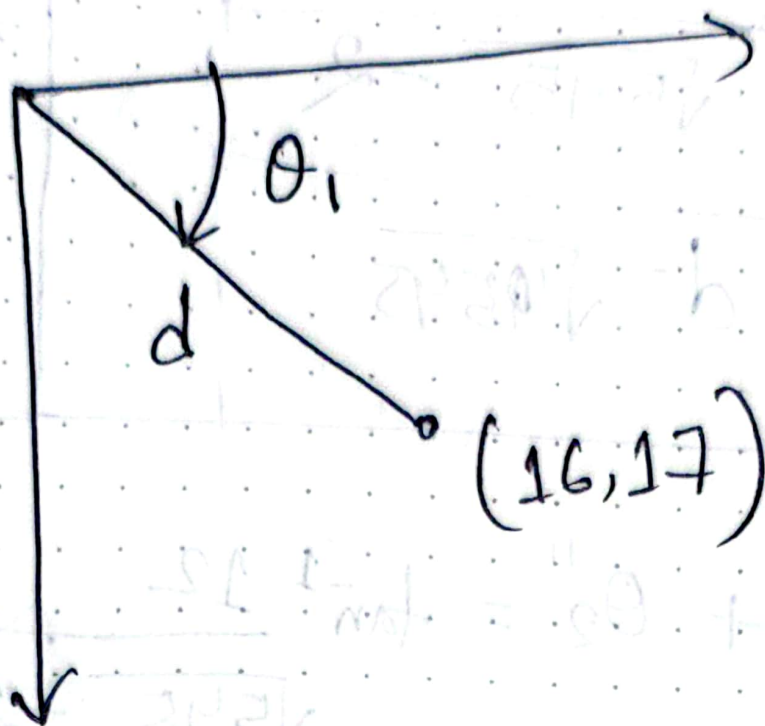
$x = 11.95$

$y = -14.23$

$z = -10$

[Ans]

Q2) Top View



$$|\theta_1| = \tan^{-1} \frac{17}{16} = 46.74^\circ$$

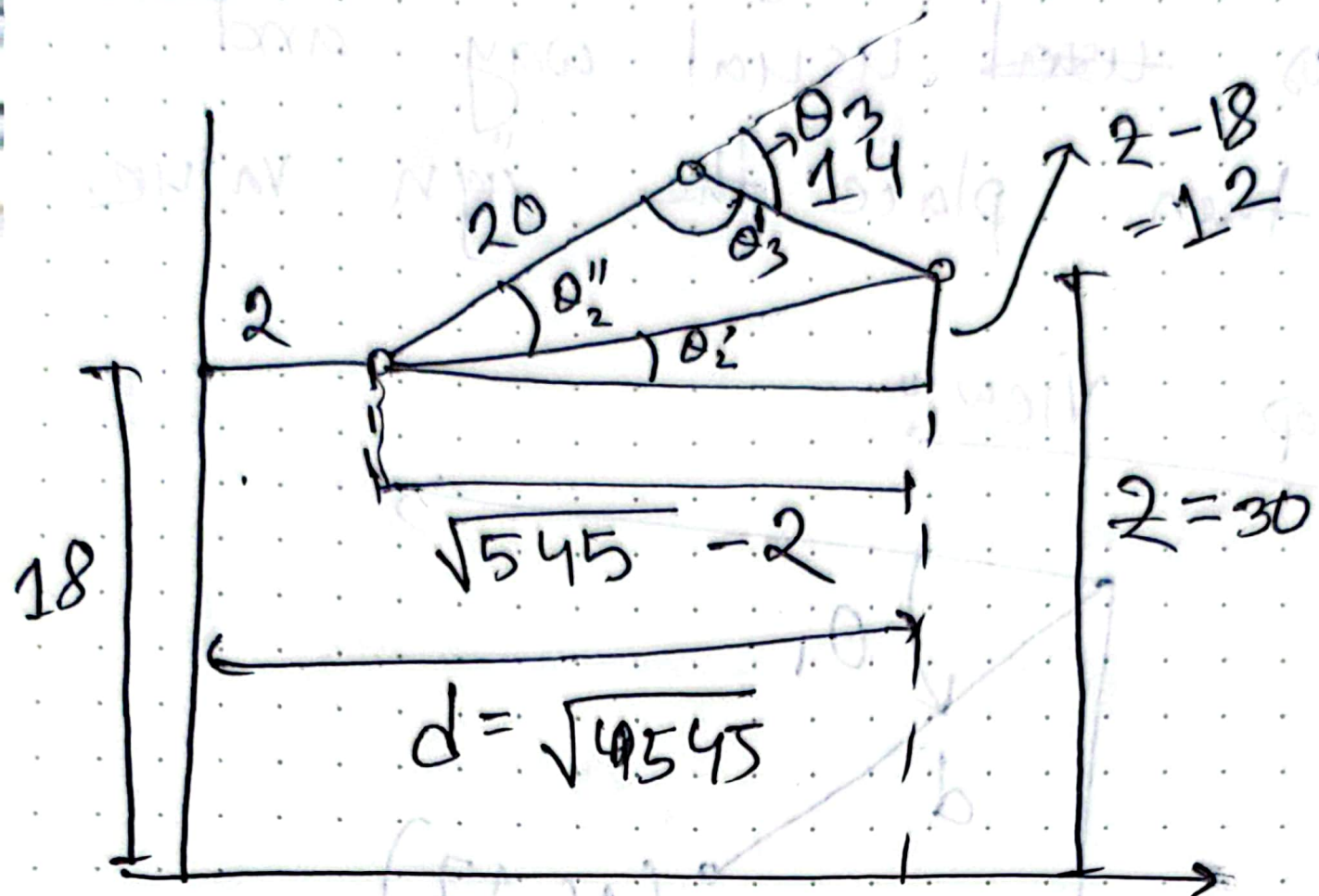
$$\therefore \theta_1 = -46.74^\circ \quad [\text{Ans}]$$

$$d = \sqrt{16^2 + 17^2} = \sqrt{545}$$

Date: _____



Side View:



$$|\theta_2| = \theta_2' + \theta_2'' = \tan^{-1} \frac{12}{\sqrt{545} - 2} + \cos^{-1} \left(\frac{20^2 + 24.5^2 - 14^2}{2 \cdot 20 \cdot 24.5} \right)$$

$$= 64.2^\circ$$

$\theta_2 = 64.2^\circ$ [As it is in the positive side]

$$\theta_3' = \cos^{-1} \left(\frac{20^2 + 14^2 - 24.5^2}{2 \cdot 20 \cdot 14} \right)$$

$$= 90.43^\circ$$

$$|\theta_3| = 180^\circ - 90.43^\circ$$

$$= 89.57^\circ$$

$$\therefore \theta_3 = -89.57^\circ$$

Ans It is in

the ~~are~~ anti-clock-wise
side